

Qatar-9b

R-2480

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-9_250131 · created 2026-08-24T00:10:51+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460706.874 +/- 0.011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.206 +/- 0.06
Transit depth (Rp/Rs)^2	4.25 +/- 2.46 [%]
Orbital Inclination (inc)	87.78 +/- 1.64
Airmass coefficient 1 (a1)	16.62 +/- 6.02
Airmass coefficient 2 (a2)	-2.5 +/- 1.02
Scatter in the residuals of the lightcurve fit is	37.84 %
Best Comparison Star	#2 - [323, 311]
Optimal Aperture	26.73
Optimal Annulus	37.03
Transit Duration (day)	0.0782 +/- 0.0057

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.206 $\pm$ 0.06 vs 0.1489 (deviation 0.8 σ)
Duration fit vs published	0.0782 d vs 0.0779 d (deviation 0.04 σ)
Mid-time O-C	14.1 min
Depth SNR	2.9
Residual scatter	37.84 %

Light curve

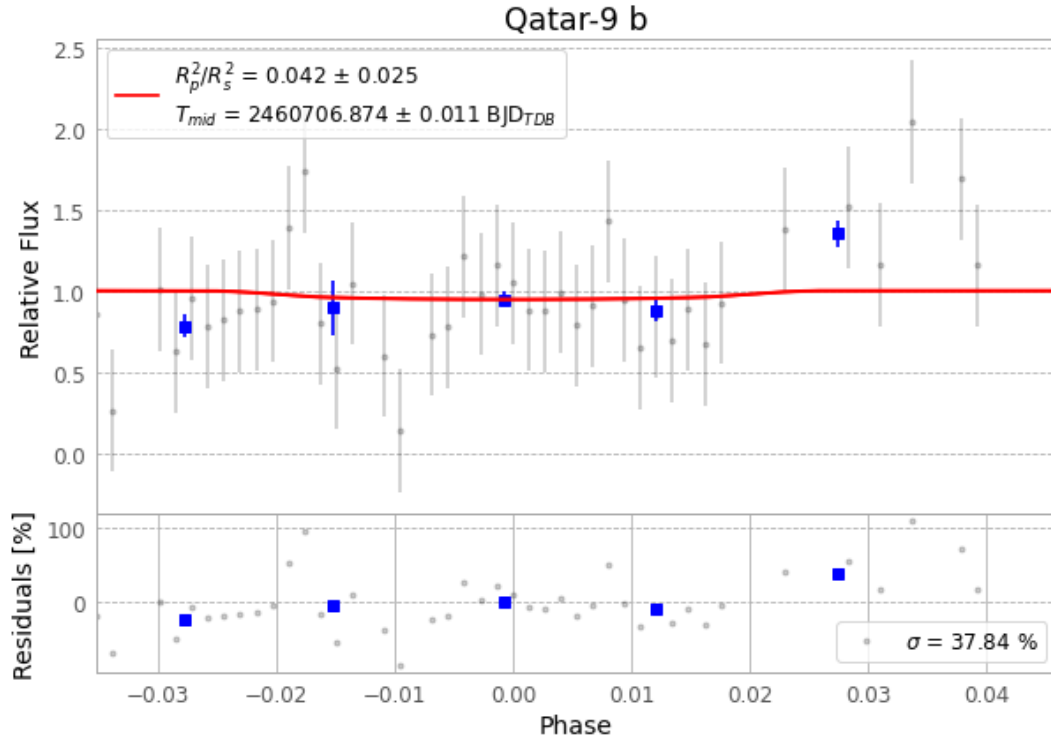


Figure 1 — FinalLightCurve_Qatar-9 b_2025-01-31.png (SHA-256 825036cab505eec2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [289, 405], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b44bab5534da8542...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

61 FITS frames (40 MB), Qatar-9250131073316.FITS → Qatar-9250131103316.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-9-250131.log (439a22670ec1...), FinalLightCurve_Qatar-9 b_2025-01-31.png (825036cab505...), FinalLightCurve_Qatar-9 b_2025-01-31.csv (28d44c350107...)

Compute resources

Wall time 272.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-24 00:10Z.